# Mahdi Haji

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## Education

- **PhD in Computer Science**, ***University of Oxford*** 2026–2029
  Working on **Reliability of AI Infrastructure** under the supervision of **Prof. Nobuko Yoshida**

- **BSc in Computer Science**, ***University of Tehran*** 2021–2025
  Worked on **Deterministic Behavior of LLMs at Inference Time** under **Prof. Mehdi Kargahi**

## Research Experience

- **Research Assistant at Ordered Systems Lab, *University of Michigan*** 2024/04–2026/05
  *Supervised by Prof. Ryan Huang* - (Remote Collaboration, Part-time)
  - Engineered a runtime validator to detect silent semantic violations in production-scale distributed systems including ZooKeeper, HBase, Kafka and Cassandra.
  - The results contributed to a paper that submitted to OSDI 2027.

- **Research Intern, *McGill University*** 2024/01–2024/04
  *Supervised by Dr. Majid Babaei*
  - Expanded a Knowledge Graph–based framework for detecting system performance anomalies using tracing data from LTTng with a Retrieval-Augmented Generation (RAG) approach.

- **Research Assistant, *Birmingham City University*** 2026
  *Supervised by Prof. Fathiyeh Faghih* - (Part-time)
  - Developed a framework for evaluating test-suite adequacy of Transformer encoder models, defining an attention "behaviour space" and new coverage criteria to target diverse model behaviours.
  - Built an online error-prediction model on this behaviour space to flag likely mispredictions at deployment time, reaching 93% AUC on BERT and 84% on ViT.

- **Research Assistant, *University of Tehran*** 2025
  *Supervised by Prof. Kargahi* - (Bachelor Project - Part-time)
  - Investigated sources of non-determinism in large language model (LLM) inference and proposed methods to ensure consistent outputs in real-time embedded cyber-physical systems.
  - Built a framework leveraging LLM non-determinism to auto-generate test cases for CPS.

## Publications

- **Chess: Test-Guided Program Synthesis for Distributed System Checkers** 2026
  Wanning He, **Mahdi Haji**, Rui Dong, Pinhan Zhao, Xinyu Wang, Peng Huang
  *Submitted to 21st USENIX Symposium on Operating Systems Design and Implementation - OSDI'27.*

- **From Attention to Assurance: Enhancing Transformer Encoder Reliability through Advanced Testing and Online Error Prediction** 2026
  Pouria Golshanrad, **Mahdi Haji**, Fathiyeh Faghih,
  *Submitted to Journal of Systems and Software (JSS), Special Issue: Bridging AI, Automation, and Sustainability in Software Testing.*

- **Argos: A Decentralized Federated System for Detection of Traffic Signs in CAVs** 2024
  **Mahdi Haji**, Alireza Hosseini, Amirali Shahriary, Soheil Hajian
  *Available as a preprint on arXiv; presented at the Flower Event,* ***University of Cambridge.***

## Selected Industry Experience

- **Lead Engineer at Enigma Investing** 2023/07–2026/10
  - Designed a portfolio management platform for fund and asset managers, computing real-time Net Asset Value (NAV) across **$8B+ in client assets**.
  - Led the development of a high-concurrency Django backend, optimizing financial processing to reduce NAV calculation time **from 3 minutes to under 2 seconds**.
  - Managed **a 7-engineer team** across system architecture, code quality, and deployment strategy.
- **Software Engineer at Rastai.ai** 2023/02–2023/07
  - Built a platform to personalise E-Learning for individual learners using Reinforcement Learning, **improving course completion rates by 42% in A/B tests.**
  - Served as the backend developer, handling the design of the platform's infrastructure, **supporting 10k concurrent requests with less than 200ms latency.**
- **Co-founder at Biturnix** 2022/01–2023/01
  - Co-founded a startup building AI tools for strategic business insights and automated customer support, growing to 10+ business clients within the first year.
  - Transitioned the product to the **accelerator program at IUST** after one year of operation.
- **Data Engineer Intern at Vasl Company** 2021/06–2022/01
  - Built data pipelines and real-time synchronization systems across multiple databases, and designed analytical dashboards to support data-driven business decisions.
  - Helped pivot the data pipeline architecture from **custom Python scripts to Apache NiFi**; built a **Change Data Capture (CDC) pipeline on MongoDB** and integrated it with Kibana.

## Technical Expertise

- **Languages & Frameworks:** Django, SpringBoot, Python, Kotlin, C++, Java, Go, SystemVerilog
- **Infrastructure:** Docker, Kubernetes, PostgreSQL, MongoDB, ELK
- **Distributed Systems:** ZooKeeper, Kafka, HBase, Consensus Protocols
- **Artificial Intelligence:** Federated Learning, ML, PyTorch, Multi-GPU/Multi-Node Training
- **Formal Methods:** Session types (MPST), Process Algebra

## Teaching & Academic Services

**Demonstrator**, University of Oxford
*Concurrent Programming, Prof. Cheval*

**Head Teaching Assistant**, UT
*Operating Systems, Prof. Kargahi*

**Head Teaching Assistant**, UT
*Data Structures & Algorithms, Prof. Faili*

**Head Teaching Assistant**, UT
*Software Testing, Dr. Khames-Panah*

**Teaching Assistant**, UT
*Compiler & PL, Dr. Tavasoli*

**Olympiad Instructor**, Rouzbeh School
*Number Theory & Geometry for IMO students*

## Honors and Awards

- Ranked 160 nationwide in the University Entrance Exam among over 300,000 participants 2021
- Galamchi's Scholarship for top rank in the University Entrance Exam (for 2 years) 2021
- Rank 1st in the Spaghetti Coding Contest at Sharif University of Technology 2025
- Ranked 3rd in cohort of 130 CS students at University of Tehran. 2025
- Exam-exempt direct admission to any MSc program in Iran as a top student (Declined) 2025